

wind power projects were completed in 2019.

Iea.org further reported that the share of renewables in global electricity generation jumped to nearly 28 per cent in Q1 2020 from 26 per cent in Q1 2019. The increase in renewables came mainly at the cost of coal and gas, though those two sources still represent close to sixty per cent of global electricity supply. In Q1 2020, variable renewables—in the form of solar PV and wind power—reached nine per cent of generation, up from eight per cent in Q1 2019.

Renewable energy is the fastest-growing energy source in the United States, increasing 100 per cent from 2000 to 2018, and it made up more than 17 per cent of net US electricity generation in 2018, with the bulk coming from hydropower (7.0 per cent) and wind power (6.6 per cent).

Around eighteen percent of the energy consumed globally for heating, power, and transportation was from renewable sources in 2017. Nearly sixty percent came from modern renewables (biomass, geothermal, solar, hydro, wind, and biofuels) and the remainder from traditional biomass (used in residential heating and cooking in developing countries). Renewables made up 26.2 per cent of global electricity generation in 2018. That's expected to rise to 45 per cent

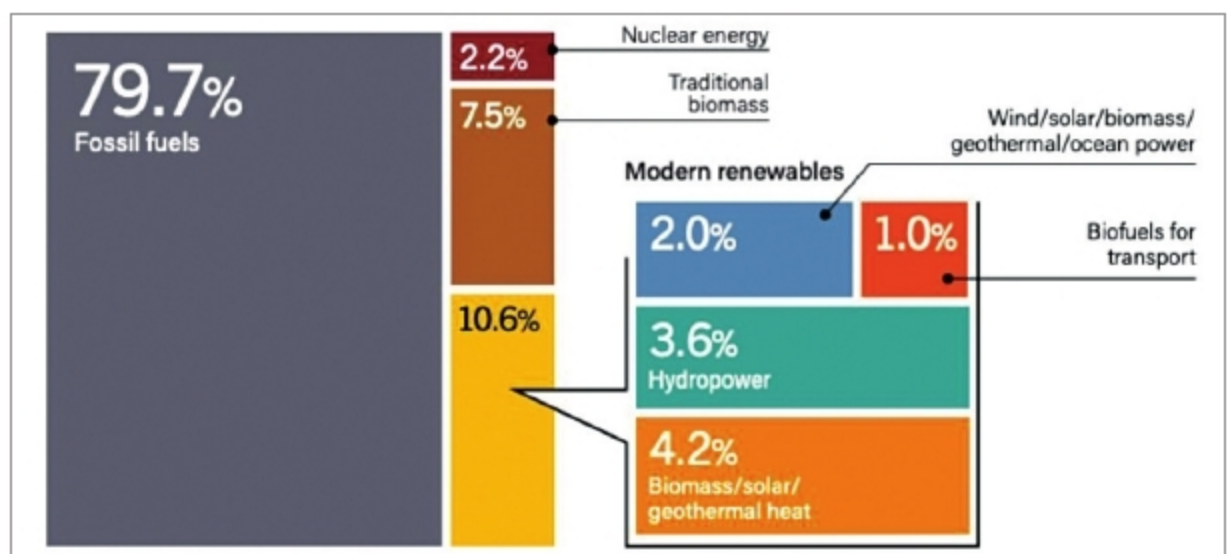


Fig. 3: Planned sources of energy and their share by 2022 (Source: Center for Climate and Energy Solutions)

by 2040. Most of the increase will likely come from solar, wind, and hydropower.

Growth of renewable energy in India

Our nation's renewable energy generation increased to 5.22 per cent in first eight months of the previous financial year (April-November 2019), the lowest rate of growth in the past four years since 2015-16, according to fresh data sourced from the Central Electricity Authority (CEA). CEA's monthly executive summary report of November 2019 showcased combined generation from solar, wind, small hydro and biomass and bagasse-based power plants stood at 93.69 billion units in the April-November period.

India now ranks fourth in wind power, fifth in solar power and fifth in renewable power installed capacity as of 2018, and it posted a CAGR of 17.33 per cent between FY 2016-2020. As a part of its Paris Agreement commitments, the government has set an ambitious target of achieving 175GW of renewable energy capacity by 2022 (Fig. 3). This includes 100GW of solar capacity addition and 60GW of wind power capacity.

Government plans to establish a renewable energy capacity of 500GW by 2030. According to the gov-

ernment trust India Brand Equity Foundation (IBEF), the installed renewable energy capacity stood at 87.26GW, of which solar and wind comprised 34.81GW and 37.74 GW, respectively, as of April 2020. In 2019, India installed 7.3GW of solar power across the country, establishing its position as the third-largest solar market in the world.

Challenges of installing renewable energy

Though the sector poses several benefits to society and industry, it also poses a lot of challenges. The solar energy segment has a bright future and holds immense potential for our economy. However, the lack of clear policies and regulations that can promote the implementation of renewable energy technologies is impeding the larger adoption in this segment. The market in renewable energies requires consistent policies and legal procedures to increase investor interest.

Lack of knowledge and understanding of renewable energy technologies and systems among rural communities is another challenge faced in the growth of renewable energy. Initial resources, implementation costs, economic status, and the availability of incentives and subsidies are the key factors influencing the pace of adoption of renewable energy technologies.

It is being witnessed that renewable energy's initial investment cost is much higher when compared to other traditional energy sources. This remains a major obstacle to sustainable renewable energy solutions being put into practice. **EFY**

Period	Figures in Billion Units	% growth
April-November FY16	46.16	
April-November FY17	58.57	26.88
April-November FY18	70.02	19.54
April-November FY19	89.04	27.16
April-November FY20	93.69	5.22

Fig. 4: Cumulative RE generation (Source: CEA)

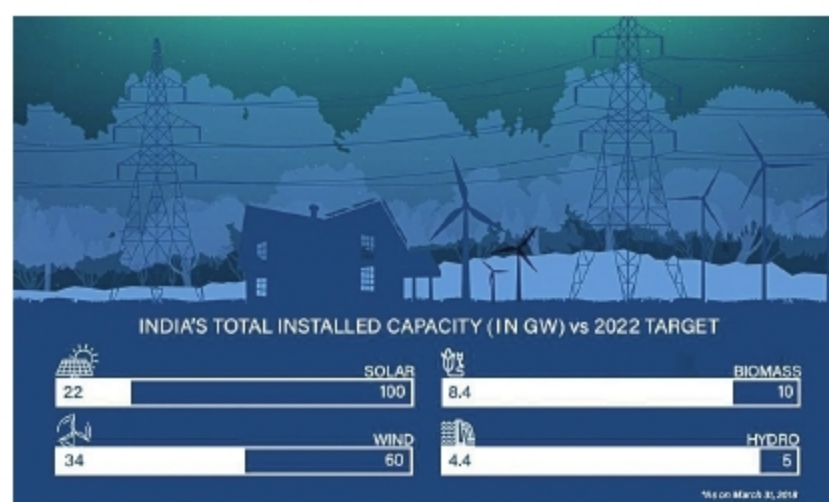


Fig. 5: India's total installed capacity in 2018 (Source: teriin.org)